Docker has a wide range of parameters and options that you can use to configure your containers, images, and the Docker daemon itself. Below is a comprehensive list of commonly used Docker parameters along with their definitions and descriptions.

Docker CLI Command Options

`docker run`

- **`-d, --detach`**
  - **Definition**: Run container in background and print container ID.
  - **Description**: Allows you to run a container in detached mode.

- **`-i, --interactive`**
  - **Definition**: Keep STDIN open even if not attached.
  - **Description**: Keeps the standard input open for the container.

- **`-t, --tty`**
  - **Definition**: Allocate a pseudo-TTY.
  - **Description**: Allocates a terminal for the container.

- **`-p, --publish`**
  - **Definition**: Publish a container's port(s) to the host.
  - **Description**: Maps a container's port to a port on the host.

- **`-e, --env`**
  - **Definition**: Set environment variables.
  - **Description**: Passes environment variables to the container.

- **`--env-file`**
  - **Definition**: Read in a file of environment variables.
  - **Description**: Loads environment variables from a file.

- **`-v, --volume`**
  - **Definition**: Bind mount a volume.
  - **Description**: Mounts a volume from the host into the container.

- **`--name`**
  - **Definition**: Assign a name to the container.
  - **Description**: Names the container for easier management and reference.

- **`--network`**
  - **Definition**: Connect a container to a network.
  - **Description**: Specifies the network the container should join.

- **`--restart`**
  - **Definition**: Restart policy to apply when a container exits.
  - **Description**: Sets the restart policy for the container (e.g., always, on-failure).

- **`--rm`**
  - **Definition**: Automatically remove the container when it exits.
  - **Description**: Cleans up the container after it stops.

- **`--cpus`**
  - **Definition**: Number of CPUs.
  - **Description**: Limits the number of CPUs the container can use.

- **`--memory`**
  - **Definition**: Memory limit.
  - **Description**: Specifies the maximum amount of memory the container can use.

- **`--entrypoint`**
  - **Definition**: Overwrite the default ENTRYPOINT of the image.

  - **Description**: Overrides the default entrypoint set by the Docker image.

- **`--detach-keys`**
  - **Definition**: Override the key sequence for detaching a container.
  - **Description**: Customizes the key sequence for detaching from the container.

`docker build`

- **`-t, --tag`**
  - **Definition**: Name and optionally a tag in the 'name:tag' format.
  - **Description**: Tags the image with a name and optionally a tag.

- **`-f, --file`**
  - **Definition**: Name of the Dockerfile.
  - **Description**: Specifies the Dockerfile to use for building the image.

- **`--build-arg`**
  - **Definition**: Set build-time variables.
  - **Description**: Passes arguments to the Docker build process.

- **`--no-cache`**
  - **Definition**: Do not use cache when building the image.
  - **Description**: Forces Docker to build the image from scratch.

- **`--target`**
  - **Definition**: Set the target build stage to build.
  - **Description**: Allows building a specific stage in a multi-stage Dockerfile.

`docker-compose`

- **`up`**
  - **Definition**: Build, create, start, and attach to containers for a service.

- **Description**: Brings up the entire application stack as defined in the `docker-compose.yml` file.

- **`down`**
  - **Definition**: Stop and remove containers, networks, images, and volumes.
  - **Description**: Takes down the application stack and cleans up resources.

- **`build`**
  - **Definition**: Build or rebuild services.
  - **Description**: Builds images for services defined in the `docker-compose.yml` file.

- **`ps`**
  - **Definition**: List containers.
  - **Description**: Lists all containers in the application stack.

- **`logs`**
  - **Definition**: View output from containers.
  - **Description**: Shows the logs of the containers.

- **`exec`**
  - **Definition**: Execute a command in a running container.
  - **Description**: Runs a command inside a running service container.

Dockerfile Instructions

- **`FROM`**
  - **Definition**: Sets the Base Image for subsequent instructions.
  - **Description**: Specifies the base image to use for the build process.

- **`RUN`**
  - **Definition**: Executes a command in a new layer on top of the current image.
  - **Description**: Runs commands to set up the image, e.g., installing software.

- **`CMD`**
  - **Definition**: Provides defaults for an executing container.
  - **Description**: Sets the default command to run when the container starts.

- **`LABEL`**
  - **Definition**: Adds metadata to an image.
  - **Description**: Provides key-value pairs of metadata for the image.

- **`EXPOSE`**
  - **Definition**: Informs Docker that the container listens on the specified network ports at runtime.
  - **Description**: Documents the ports the container will use.

- **`ENV`**
  - **Definition**: Sets environment variables.
  - **Description**: Specifies environment variables to set in the container.

- **`ADD`**
  - **Definition**: Copies new files, directories, or remote file URLs from <src> and adds them to the filesystem of the container at the path <dest>.
  - **Description**: Adds files from the host system or URLs to the container.

- **`COPY`**
  - **Definition**: Copies new files or directories from <src> and adds them to the filesystem of the container at the path <dest>.
  - **Description**: Similar to `ADD`, but without the additional capabilities (e.g., no remote URLs).

- **`ENTRYPOINT`**
  - **Definition**: Configures a container that will run as an executable.

- **Description**: Sets the main command for the container, which cannot be overridden at runtime.

- **`VOLUME`**

  - **Definition**: Creates a mount point with the specified path and marks it as holding externally mounted volumes from native host or other containers.

  - **Description**: Defines a directory in the container that can be mounted from the host.

- **`USER`**

  - **Definition**: Sets the user name or UID to use when running the image and for any `RUN`, `CMD`, and `ENTRYPOINT` instructions.

  - **Description**: Specifies the user to use for running commands.

- **`WORKDIR`**

  - **Definition**: Sets the working directory for any `RUN`, `CMD`, `ENTRYPOINT`, `COPY`, and `ADD` instructions.

  - **Description**: Changes the working directory for subsequent instructions.

- **`ARG`**

  - **Definition**: Defines a variable that users can pass at build-time to the builder with the `docker build` command.

  - **Description**: Allows you to define build-time variables.

- **`ONBUILD`**

  - **Definition**: Adds a trigger instruction to the image that will be executed when the image is used as the base for another build.

  - **Description**: Executes a command in child images.

Docker Daemon Configuration

- **`--log-level`**

  - **Definition**: Set the logging level (debug, info, warn, error, fatal).

  - **Description**: Configures the verbosity of Docker logs.

- **`--storage-driver`**
  - **Definition**: Storage driver to use.
  - **Description**: Specifies the storage driver (e.g., overlay2, aufs).

- **`--insecure-registry`**
  - **Definition**: Enable insecure registry communication.
  - **Description**: Allows communication with insecure registries.

- **`--registry-mirror`**
  - **Definition**: Preferred Docker registry mirror.
  - **Description**: Configures a mirror for Docker Hub.

- **`--default-ulimit`**
  - **Definition**: Set default ulimit options.
  - **Description**: Configures default ulimit settings for containers.

This list provides an overview of common Docker parameters and options. For a complete and detailed list, always refer to the official [Docker documentation](https://docs.docker.com/).